

Following Paper ID and Roll No. to be filled in your Answer Book.

PAPER ID : 43208Roll
No.

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B. Tech Examination 2026**(Even Semester)****PROGRAMMING CONCEPTS WITH PYTHON*****Time : Three Hours]******[Maximum Marks : 60*****Note :-** Attempt all questions.**SECTION - A**

1. Attempt all parts of the following : $8 \times 1 = 8$
- (a) What are the main steps in the python programming cycle?
 - (b) What is operator precedence in python?
 - (c) What is boolean expression?
 - (d) What is the purpose of the if else statement in python?
 - (e) What is the time complexity of the sieve of eratosthenes algorithm?

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- (f) What is the purpose of the `__init__`-method in class?
- (g) What is the purpose of an ADT interface in python?
- (h) What is method overriding in inheritance?

SECTION-B

2. Attempt any two part of the following: $6 \times 2 = 12$
- (a) Describe string operations in python. Explain length, concatenation, repetition and slicing.
 - (b) Explain the sieve of eratosthenes algorithm, ADT and inheritance in python.
 - (c) Describe list data structure in python and write a program to find sum of elements in a list.
 - (d) Explain file handling in python and write a program to write data into a file.

SECTION-C

Note:- Attempt any two parts from each question .
All part carry equal marks. $8 \times 5 = 40$

- 3. (a) Explain conditional statement in python. Also explain while loop and for loop with example.
- (b) Explain arithmetic operators, operator precedence and boolean expressions with example.
- (c) Explain functions in python with suitable examples and also discuss default and keyword arguments.
- 4. (a) Describe string operations in python. Explain different string methods: `lower()`, `upper()`, `strip()`, `replace()`, with example.
- (b) What are file handling different modes? Also explain `read()`, `readline()`, methods with example.
- (c) What is exception in python? Explain try, except, else and finally blocks.
- 5. (a) Explain how to import and use module. Also explain pandas and numpy with example.

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- (b) Define class and object in python and explain object oriented programming concept with encapsulation and abstraction.
 - (c) What is inheritance in python? Explain different types of inheritance with example.
6. (a) Explain numeric data types in python (int, float, complex) and describe type conversion between them.
- (b) Compare list, tuple, set and dictionary data types in python with respect to mutability and ordering.
 - (c) Explain detail the concept of object oriented programming (oop) in python and discuss its key principles, advantages, real world applications with suitable examples.
